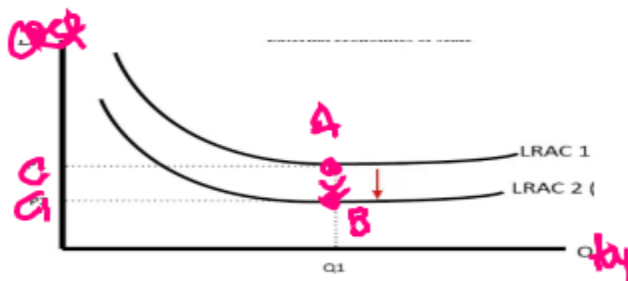


1-D:

A – Distribution of all the current profit among the existing shareholders will not bring gains from dynamic efficiency. This is because, this means that there are no retained profits available to spend on Research and Development and innovation.

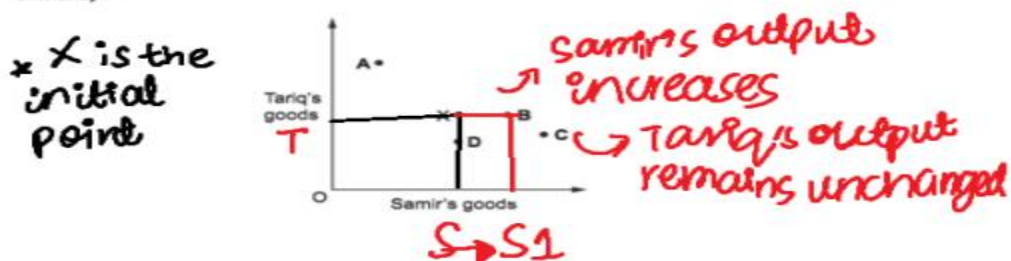
B and C : Maximizing the labor productivity of its current workers, and minimizing the average costs of production will reduce the short-run and long-run average costs of production, enabling to achieve “productive efficiency” (at a given point in time)

D – When due to the growth of the industry, firms enjoy super-normal profits, and retain them to invest in new technology, or spend on research and development which helps them to innovate and develop new processes, they are able to reduce their Long-Run Average costs “over time” (which represents “dynamic efficiency”). As a result of which, the Long-run Average cost curve shifts downwards.



2-B:

According to the Pareto criterion, which point would definitely indicate increased allocative efficiency?



→ The Pareto criterion states that a resource allocation or an economic situation is considered Pareto efficient if it is impossible to make any individual better off without making someone else worse off. In other words, a “Pareto improvement” occurs when a change in the allocation of resources results in at least one person being better off, without making anyone else worse off.

- A- Here, apart from a rise in the output of Tariq, there is a fall in the output of Samir. This means that “Pareto efficiency” is not being achieved.
- B- This is correct because here an increase in the output of Samir is occurring, along with which, the output of Tariq is unchanged.
- C- Here, apart from a rise in the output of Samir, there is a fall in the output of Tariq. This means that “Pareto efficiency” is not being achieved.
- D- Here, the output of Samir is unchanged, and the output of Tariq is shown to decrease.

3-B:

B - Merit goods are goods or services that are considered beneficial for individuals and society, but people may under-consume them due to lack of information. Subsidies for merit goods are a form of government intervention to encourage their consumption and improve overall welfare. **This option rejects the assumption that each individual is the best judge of their own economic welfare because it suggests that individuals might not fully understand the benefits of consuming certain merit goods due to lack of adequate, appropriate or accurate information about the true value these goods present, or in other words, imperfections.** Subsidies for merit goods involve the government actively intervening to influence individual consumption choices, suggesting that individuals may not always be the best judges of their own economic welfare, especially when it comes to under-consumption of beneficial goods due to information failure. This is why, key B, is correct.

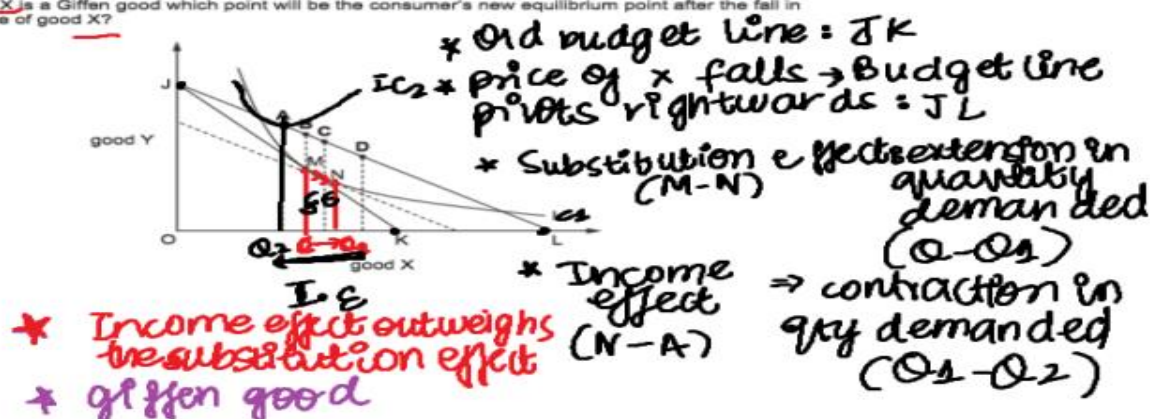
A) Pollution Controls: Pollution controls aim to address negative externalities caused by pollution and do not reject the assumption that individuals are the best judges of their own welfare.

C) The Provision of Public Goods: The provision of public goods is a government intervention to address the market failure associated with non-provision of public goods by the private sector due to the "free-rider problem". It addresses a specific market failure related to non-excludability and non-rivalry in the consumption of public goods.

D) The Regulation of Monopolies: The regulation of monopolies is aimed at ensuring fair competition and protecting consumers from potential abuse of market power by monopoly firms. This option does not reject the assumption of individuals being the best judges of their own economic welfare.

4-A:

4. In the indifference curve diagram point M is the consumer's initial equilibrium and MN is the substitution effect of a fall in the price of good X.
If good X is a Giffen good which point will be the consumer's new equilibrium point after the fall in the price of good X?



Since income effect outweighs the substitution effect, the following good is said to be a "Giffen good". In case of Giffen goods, there is a positive relationship between price and quantity demanded. If for example, the price of a loaf of bread falls, people will contract some quantity of loaves to purchase some meat which they can now afford, in order to feel richer.

5-D:

In order to maximize utility, the utility-maximizing condition or the equi-marginal principle must be established.

$$\frac{\text{Marginal utility } x}{\text{Price } x} = \frac{\text{Marginal utility } y}{\text{Price } y}$$
$$\frac{\text{Marginal utility } x}{\text{Marginal utility } y} = \frac{\text{Price } x}{\text{Price } y}$$

6-C:

A - This statement is the correct one because a **"kinked demand curve"** represents the individual firm's expectations about other firm's responses to changes in its prices. -If for example, the firm raises its price, no other firm in the industry will follow, and therefore, this represents **"elastic demand"** (because, it is the behavior of a firm), and as a result, there is a contraction in quantity demanded by a relatively higher percentage, which results in a **"fall in Total Revenue"**.

-Importantly, when a firm decreases its price, all other firms in the industry will follow, and therefore, this represents **"inelastic demand"** (because, it is the behavior of an industry), and as a result, there will be an extension in qty demanded by a relatively smaller percentage, which results in a **"fall in Total Revenue"**

Therefore, prices remain rigid or unchanged at the "kink"

B – This is also the correct statement because the marginal revenue curve has a vertical segment at the market price, or in other words, the **"marginal revenue curve is discontinuous"**.

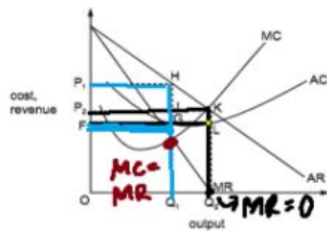
C – This statement is incorrect, because of which, **this option becomes correct.**

The **kinked demand curve model** of oligopoly **does not determine** how the **equilibrium market price is determined**. Instead, it **only explains why the "equilibrium price is rigid"**. This is why, **key C, will be correct.**

The kinked demand curve model provides insights into the pricing behavior of individual firms in an oligopoly and how they might avoid significant price competition through price stickiness within a certain range of marginal cost.

D - The kinked demand curve model suggests that **firms are likely to keep their prices stable within a certain range of marginal costs**, inside of the **operational range (which represents the range whereby Marginal Revenue is discontinuous)**, which **corresponds to the "price at the kink"**. This is because changes in price outside this range will result in a fall in total revenues.

7-C:



When $MR=0 \rightarrow$ sales revenue maximization

* $Costs = OFLQ_2$

* $Revenue = OP_2KQ_2$

* Profit: P_2KLF

Final profit

Final: * When $MC = MR \rightarrow$ profit-maximization

* $Costs \rightarrow OFGQ_1$

* $Revenue \rightarrow OP_1HQ_1$

* Profit $\rightarrow P_1HGF$

Original profit

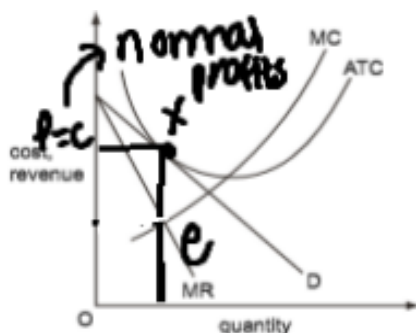
8-B:

A – In case of price-elastic demand, whereby a smaller percentage change in price brings about a larger percentage change in qty demanded, a fall in price, since brings about an extension in qty demanded by a relatively higher percentage, will result in a rise in "Total revenue". This is why, this is incorrect.

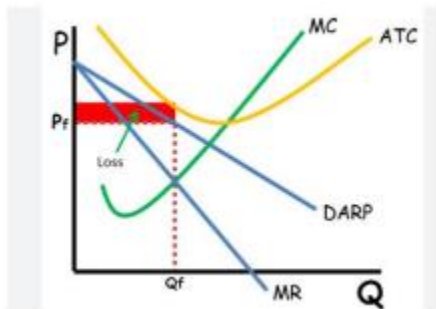
B – When the demand for a product is price-inelastic, a fall in price will bring about an extension in qty demanded by a relatively smaller percentage, leading to a "fall in Total Revenue". Here, since an increase in output, is shown to result in a fall in Total Revenue, this becomes correct.

C and D: This describes a concept associated with demand ; therefore, we are interested in information about price elasticity of demand, instead of "price-elasticity of supply"

9-A:

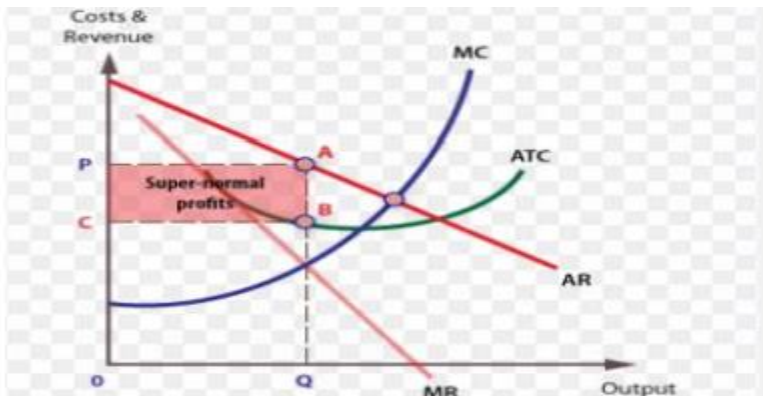


A – This is correct because this diagram represents “long-run normal profits”. This is because, at the profit-maximizing level of output, Average revenue is equal to Average cost, which represents a situation of “normal profits” or “zero economic profits”



B - A firm in a monopolistically competitive firm will make short-term losses or subnormal profits when Average revenue is lesser than Average cost. This can be seen in the diagram above.

C – When a firm in perfect competition is in long-run, the demand curve will need to be horizontal because Average revenue=marginal revenue. This represents a scenario whereby the Average revenue is not equal to marginal revenue, which shows “imperfect competition”



D - A monopoly makes a supernormal profit when Average revenue is greater than Average cost. This can be seen above. Importantly, the demand curve of a monopoly is inelastic, and therefore, its Marginal revenue curve goes to negative.

10-B

A – A greater bargaining power in purchasing supplies represents “purchasing economies of scale”

B- Risk bearing economies of scale creates room for business to spread risks on different products that a company can fall back on. Having different products helps to diversify or expand the product range of the business that enables the business to achieve “**Risk-bearing economies of scale**”

C- Lower costs in raising capital explains “financial economies of scale”

D – Lower distribution costs by increasing market share represents “selling or marketing economies of scale”

➔ High-growth firms can leverage economies of scale in their marketing and distribution activities. They can spread their advertising, promotion, and distribution costs over a larger customer base, reducing the cost per customer reached. Additionally, they may be able to negotiate better contracts with logistics providers or achieve more favorable terms with retailers, further reducing costs.

11-C:

A - This option is incorrect. While maximizing output could be a strategy to gain a larger market share, it is not related to eliminating competition and becoming a monopoly. In some cases, maximizing output could lead to increased competition. In an oligopoly, a few dominant firms control the market. If one firm in the oligopoly decides to maximize output, it can lead to a competitive reaction from other firms. Competitors may match the increase in output to maintain their market share and pricing power. This mutual response could result in a price war or increased advertising efforts to capture consumers' attention.

B – If a firm raises its prices to maximize profits, it will emerge uncompetitive, as a result of which, its demand will fall, and therefore, it will further lose its market share.

C - This option is correct because one of the strategies that a firm can adopt to establish a monopoly position is to engage in predatory pricing. Predatory pricing involves setting prices below the cost of production or significantly lower than competitors' prices to drive them out of the market. By reducing prices below their rivals' cost levels, the firm aims to create financial difficulties for other competitors, forcing them to exit the market due to sustained losses. Once competitors exit, the firm can then raise prices and dominate the market as a monopoly.

D- Even if the number of its suppliers are reduced, the material will still be available, and other firms can still produce the product. And also, it is difficult to control supplies or reduce suppliers.

12-A:

A) Small design firms need to be flexible to cope with frequent fashion changes, as they need to constantly update their designs to keep up with the latest trends. On the other hand, large retail firms prioritize exploiting marketing economies of scale to maximize their reach and sales. This explains why key A will be correct.

B) This option doesn't quite fit, as small design firms may not necessarily need to employ highly specialized and skilled workers, and large retail firms typically operate at a high minimum efficient scale, not low.

C) This option doesn't quite fit, as small design firms typically don't need to operate at a high minimum efficient scale. However, large retail firms may need to offer a wide range of products to stay competitive.

D) This option doesn't quite fit, as small design firms may not necessarily need to overcome high barriers to entry, and large retail firms may not necessarily need to take advantage of technical economies of scale. (Instead, large manufacturing firms may need to take advantage of technical economies of scale)

13- C:

A - Contestable markets refer to markets with low entry and exit barriers, allowing new firms to enter and compete with existing firms easily. As a result of which, there is a threat of “potential competition” and because of which, firms behave as if there was actual competition.

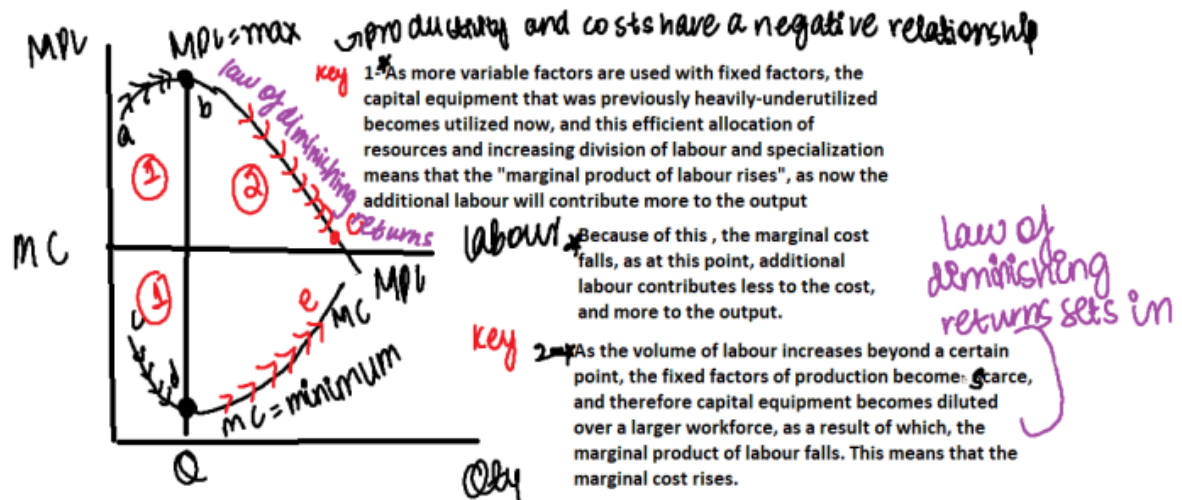
B - Diseconomies of scale occur when a firm's average costs increase as the firm's scale of production grows. This is not linked to the “separation of ownership and control” in a firm

C - The principal-agent problem arises when there is a separation of ownership and control in a firm. The owners (principal) of the firm delegate decision-making authority to managers (agents) to run the day-to-day operations and make business decisions on their behalf. However, the interests of the owners and the managers may not always align, leading to conflicts of interest.

Managers might pursue personal goals or prioritize their own interests over those of the shareholders. They may make decisions that maximize their own benefits, such as increasing their salaries and bonuses by pursuing the objective of "sales revenue maximization" because managerial salaries and bonuses are linked to revenues as compared to profits, even if these decisions do not align with the long-term interests of the shareholders, who prefer "profit maximization". As a result, the firm's performance and value may not be maximized, and the owners may experience a reduction in their wealth. This is why, key C, becomes correct.

D - The prisoner's dilemma is a game theory concept that is related to "duopoly" It is not linked to separation of ownership and control in a firm.

14-D:



In this question, we are told that the additional labor is contributing less to the total product, this is because the marginal product of labor falls. A fall in marginal product of labor explains the law of diminishing returns. This can be seen above.

A – When marginal product of labor rises, that is when, the marginal costs decrease. This can be seen above

B- Diseconomies of scale is associated with a rise in the Long Run Average Cost.

C – Increasing returns to scale is associated with a fall in the Long-Run Average Cost.

D- It can be seen above in key 2 that as the marginal product of labor falls, the "law of diminishing returns" sets in. This is why, key D, becomes correct.

15-B:

* Marginal cost $\Rightarrow$
 $\hookrightarrow$ Change in ^{of 11th} total cost
 $\hookrightarrow$ Total cost of 11th worker - Total cost of 10th worker
 $\hookrightarrow (11 \times 8.5) - (10 \times 8)$
 $MC = \$13.50$

16-C:

Marginal revenue productivity of labor at any given level of employment is equal to the Marginal Physical Product of Labor multiplied by the Marginal Revenue, and so because of this formula, only key C, becomes correct. All other options therefore, become incorrect.

17-D:

Progressive income taxes are a policy where higher income earners are taxed at a higher rate than lower income earners. This is because, they are progressive in nature. This reduces the disposable income of higher income earners and increases the disposable income of lower income earners, leading to a more equal distribution of disposable incomes between households.

A) Government support for trade unions can lead to higher wages for unionized workers, but may not necessarily lead to a more equal distribution of **disposable incomes** between all households.

B) Import duties on manufactured goods represent indirect taxes which are regressive in nature.

C) Minimum wage policy can help increase the incomes of low-wage workers, but may not necessarily lead to a more equal distribution of **disposable incomes** between all households.

18-A:

A- When the price (average revenue) is equal to the marginal cost, this is when allocative efficiency is achieved. This is why, this becomes correct.

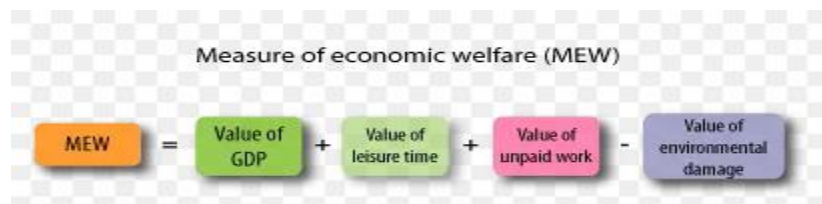
B- If the government sets the price where the marginal cost is below average cost, this means that the average cost is not minimized (because average cost is at its minimum when the marginal cost curve intersects the average cost curve, or in other words, marginal cost is equal to average cost). As a result of which, productive efficiency is not being achieved.

C- Producer surplus increases upon a rise in price. This means that, if the intervention results in a rise in producer surplus, firms will be benefitted and therefore, efficiency will reduce. **The aim of allocative efficiency is to maximize the total welfare of the society and not only the welfare of the producers.**

D- "Predatory pricing" is when prices are charged below cost to destroy competition. This when reduces competitive pressures, results in X-inefficiency.

19-D:

The "Measure of Economic Welfare" (MEW) is an alternative measure of economic well-being that adjusts GNP to take account of factors like leisure time, non-marketed activities, environmental damage and crime-rate.



If the value of non-marketed activities and leisure is high, it would cause the MEW to be greater than the Gross National Product (GNP). This is because, in that case, MEW will be: GNP + value of leisure and un-marketed activities.

A) Negative externalities like pollution would actually decrease the MEW, as they have a negative impact on economic well-being. $\text{MEW} = \text{GNP} - \text{environmental damage}$.

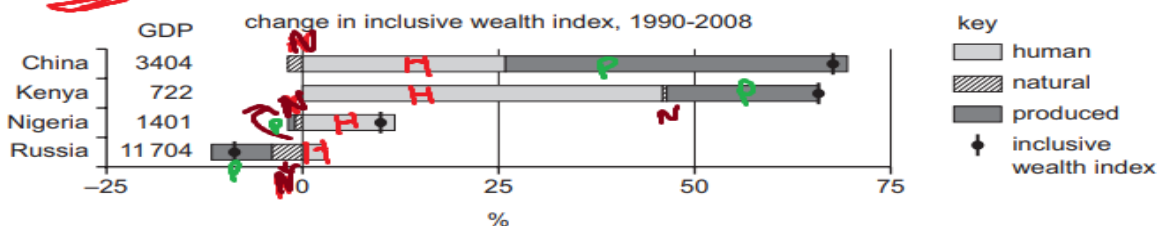
B) Property income received from abroad is already included in GNP, so it wouldn't cause MEW to be greater.

C) Regrettable necessities will also be already included in the GNP.

20-D:

20 In 2012 a United Nations report calculated the stock of wealth of 20 countries in terms of human, natural and produced resources. This was measured as the Inclusive Wealth Index (IWI).

The diagram shows the annual percentage (%) change in the IWI between 1990 and 2008 of the economies with the fastest and the slowest growth in IWI. It also shows their 2008 GDP per head (\$).



A – This is incorrect because although Kenya had the lowest level of GDP per head out of the countries mentioned, it was able to build stocks of wealth, as it is seen to experience an increase in human, natural and produced resources all at the same time. Additionally, its inclusive wealth index is the second highest.

B – This is incorrect because Kenya was able to prevent depletion of natural resources as it is experiencing an increase in natural resources.

C – This is incorrect because although Kenya had a fast growth in IWI, it still has a low GDP per head.

D – This is correct because it can be seen that all four countries are experiencing an increase in human resources.

21-D:

In less developed economies, it is common to find:

- Low saving ratios: Due to lower incomes, people in less developed economies tend to save less.
- High population growth rates: Less developed economies often have higher population growth rates due to factors like limited access to family planning and lower education levels.

This explains why key D is correct.

A) Less developed economies often have lower capital: output ratios due to limited capital stock and reliance on labor intensive techniques. Additionally, they are known to have higher external debt.

B) Less developed economies have higher external debt and a lower capital : output ratio.

C) This is incorrect because less developed economies have a lower saving ratio, and they usually have a higher population growth.

22-B:

Investing in human capital (e.g., education, training, health-care) is likely to increase both economic growth and economic development in the long run. A more skilled and productive workforce can lead to:

- Increased economic growth: As a more productive workforce can produce more goods and services, leading to increased economic output.
- Increased economic development: Improvements in education and health are directly linked to improvements in economic development.

This explains why key B is correct.

A) A decrease in the saving ratio reduces the amount of resources available for investment, potentially reducing long-term economic growth and economic development.

C) The depletion of non-renewable resources can lead to resource scarcity, increased costs because of which when Aggregate supply reduces, this results in reduced economic growth. Additionally, resultant cost push inflationary pressures can erode the purchasing power of money, which will also hamper living standards.

D) Greater use of compulsory overtime working of labor can lead to decreased worker productivity, increased costs, and reduced economic growth in the long-term. This will also translate into reduced economic development.

23-A:

- A) Cyclical unemployment occurs when aggregate demand is deficient, meaning that there is not enough demand for goods and services in the economy to employ all available workers. This type of unemployment is caused by fluctuations in economic activity, typically during recessions or depressions.

AD falls-> production falls-> demand for labor falls-> cyclical unemployment increases.

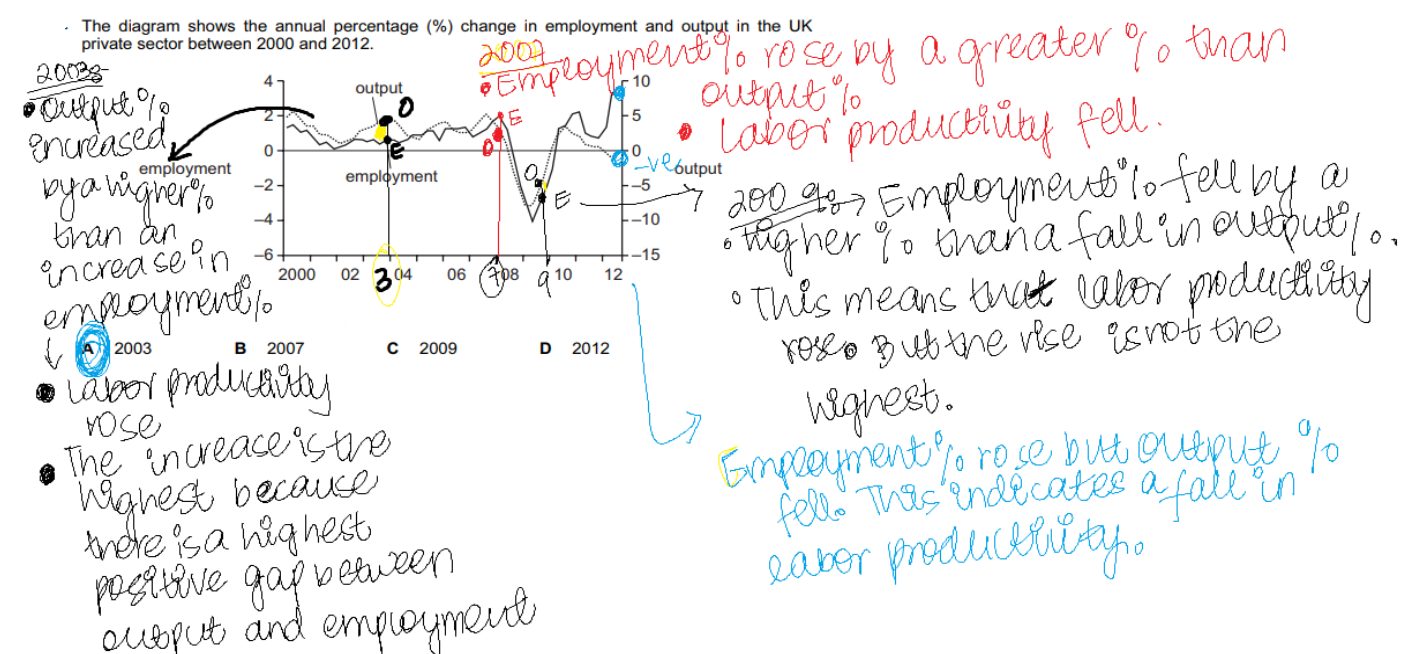
B) Regional unemployment refers to unemployment that occurs in specific regions or industries, typically due to structural factors. It is a type of structural unemployment.

C) Seasonal unemployment occurs when industries or occupations are affected by seasonal patterns, such as winter resorts or summer agriculture.

D) Structural unemployment arises from mismatches between the skills and qualifications of workers and the requirements of available jobs, often due to technological changes or shifts in industry demand or changes in the pattern of supply.

24-A:

The formula of labor productivity is: Output/Employment .



This explains why key A will be correct.

25-A:

According to Keynesian theory, an increase in the money supply will leave the level of output unchanged when the liquidity trap is operative. This occurs when interest rates are already very low, and people and businesses hoard cash even when more money is available. This might be because they might be pessimistic about future economic prospects. In this situation, the increased money supply does not stimulate economic activity.

MS increases-> but when there is a liquidity trap situation, money demand is perfectly elastic-> therefore interest rates remain unchanged-> savings, consumption and investment remains the same-> AD remains the same-> output remains the same. **This can be seen in the diagram below.**

B) If the money supply increase was not anticipated, it could still lead to increased economic activity as people and businesses will now respond to low interest rates arising as a result of an increase in money supply.

C) A floating exchange rate is a monetary regime, and an increase in the money supply would still affect output.

D) Rational Expectations theory suggests that people form their expectations about future economic outcomes based on all available information, including past experiences and current events. If an increase in money supply is announced, people may immediately adjust their expectations about future price levels, anticipating higher inflation in the future.

This means that they will increase consumption today to avoid paying higher prices in the future for the same goods. This will stimulate economic activity in the short-run and also translate into a higher output level being produced.



26-B:

The correct answer is B: $C + S + T$.

In a 4-sector economy, the equilibrium level of national income (Y) is where:

$AE=Y$.

Or in other words,

C (Consumption) + I (Investment) + G (Government Spending) + $(X - M)$ (Net Exports) = Y

For an equilibrium to exist, Y must include:

- C (Consumption): Household spending on goods and services
- S (Savings): Household savings, which represents the portion of income not spent on consumption
- T (Taxes): Government revenue from taxation, which represents the portion of income transferred to the government

Note: Y must not include imports because imports are already in the equation.

27-B:

The correct answer is B: The government keeps the unemployment benefit rate and direct and indirect tax rates unchanged.

During an economic recession, a government's budget deficit may increase due to Automatic fiscal stabilizers.

- Unemployment benefits and other social welfare programs increase as more people become eligible, leading to higher government spending.
- Lower economic activity leads to decreased tax revenues from income taxes, sales taxes, and other sources as less incomes are being earned, and there is less spending being done.

When tax revenues fall relative to government spending which automatically increases, there will be an increase in the **budget deficit of the government**.

A) Increasing tariffs on imports with inelastic demand may increase government revenue, and will not lead to an increase in the budget deficit.

C) Reducing foreign aid represents that government spending falls and widening the tax base may increase tax revenues. This will either lead to a decrease in budget deficit or an increase in budget surplus.

D) Reducing the unemployment benefit rate and decreasing the tax-free allowance on income tax would likely decrease government spending and increase revenue, respectively. This will either lead to a decrease in budget deficit or an increase in budget surplus.

28-D:

When unemployment is increasing in an economy, the tax burden on employed workers is likely to rise because:

- Government spending on unemployment benefits and other social welfare programs increases, and to finance these increased expenditures, the government may increase taxes on employed workers, leading to a higher tax burden on employed workers. **This explains why key D is likely to be correct.**

A) The human capital of unemployed workers may actually decrease due to erosion of skills and loss of work experience. **This represents the hysteresis effect.**

B) The living standards of all workers, including employed and unemployed, may decrease. The unemployed will have reduced incomes and purchasing power, whereas, the employed may face a higher tax burden, which may reduce their disposable incomes.

C) The nominal money wage rate of employed workers may fall in event of unemployment which represents a surplus of workers.

29-D:

Option A: Foreign direct investment (FDI) in developing economies can actually improve trade balances by increasing exports and reducing reliance on imports.

Option B: FDI may not necessarily reduce migration to urban areas. While FDI can create new job opportunities in urban areas, it may also lead to increased urbanization as workers move to cities to take advantage of these new opportunities.

Option C: FDI typically involves the transfer of technology to developing economies, which can enhance their productive capacity.

Option D: FDI can lead to increased economic growth and development. This is because, an increase in autonomous injections into the circular flow will result in an increase in Aggregate demand, because of which, there will be a multiplier times increase in real national income. As FDI increases income levels, consumers in developing economies can afford to buy more goods and services, leading to higher per capita levels of consumption.

30-C:

The formula of multiplier effect is: $1/(1-mpc)$ or $1/mps+mpt+mpm$

Note: By increasing the marginal propensity to consume, more of the increased government spending is circulated in the economy, leading to a larger multiplier effect and a greater impact on national income.

-> The multiplier effect of an increase in government spending on national income is influenced by the marginal propensity to consume (MPC), which is the proportion of additional income that is spent on consumption. A higher MPC means that more of the increased income as a result of an increase in government spending is spent, leading to a larger multiplier effect.

A) An increase in direct taxation would bring about an increase in marginal propensity to tax. Since this represents an increase in withdrawal from the circular flow, there will subsequently be a decrease in the marginal propensity to consume, which will bring about a fall in the value of the multiplier.

B) An increase in interest rates would reduce borrowing and subsequently consumption will fall. This can bring about a decrease in the multiplier effect.

D) An increase in the marginal propensity to import would reduce the multiplier effect, as more of the increased income is spent on imports, rather than domestic goods and services.

